
Project 1: Comparative Evaluation of Large Language Models for Beginner-Focused Bitcoin Education

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Project Overview

This project evaluates the quality of responses generated by multiple AI language models using a standardized prompt. The objective is to compare how effectively each model explains a complex topic to a beginner while following specific instructions. The evaluation focuses on clarity, accuracy, simplicity, instruction adherence, and overall usefulness.

Objective

To compare the performance of different AI language models in explaining Bitcoin to a beginner using a consistent evaluation framework.

Methodology

A standardized prompt was submitted to multiple AI language models under the same conditions. Each response was evaluated using predefined criteria: Accuracy, Clarity, Simplicity, Instruction Following, and Overall Quality. Scores were assigned based on how effectively each model explained Bitcoin to a beginner while complying with the prompt's requirements. The findings were then analyzed to identify each model's strengths, weaknesses, and overall suitability for beginner-focused educational tasks.

Prompt Used

"Explain Bitcoin to someone who has never heard of cryptocurrency. Keep the explanation under 300 words, avoid technical jargon, and include one real-world analogy."

AI Models Evaluated

ChatGPT

Claude

Gemini

Perplexity

GitHub Copilot

Evaluation Criteria

Accuracy

Clarity

Simplicity

Instruction Following

Overall Quality

Evaluation Results

Model	Accuracy	Clarity	Simplicity	Followed instructions	Overall
Perplexity	10/10	9/10	10/10	10/10	9.8/10
Claude	10/10	10/10	9/10	8/10 (slightly long)	9.5/10
Gemini	10/10	10/10	10/10	9/10	9.7/10
GitHub Copilot	2/10	8/10	8/10	0/10	4.5/10

Evaluation Scoring Framework

Each criterion was scored from 1–10 based on observable adherence to the evaluation requirements. Accuracy assessed factual correctness; clarity assessed ease of understanding; simplicity assessed avoidance of unnecessary technical language; instruction following assessed compliance with the prompt; and overall quality reflected the usefulness of the response for the intended beginner audience.

Individual Analysis

Strengths & Weaknesses

Perplexity

Strengths

Extremely beginner-friendly.

Great analogy (shared notebook).

Included practical advice about wallets and price volatility.

Stayed focused on the prompt.

Weakness

Slightly more detailed than necessary in the wallet section.

Claude

Strengths

Excellent storytelling.

Easy-to-understand explanation.

Strong real-world analogy.

Very natural writing style.

Weakness

Included information about mining, which added complexity and made the response a bit longer than requested.

Gemini

Strengths

Very well organized with headings.

Excellent explanation.

Strong analogy.

Easy for beginners to follow.

Weakness

The markdown headings make it look more like an article than a conversational explanation.

GitHub Copilot

Strengths

Correctly recognized it wasn't the best tool for the task.

Redirected the user to appropriate resources.

Weakness

Did not answer the prompt, making it unsuitable for this evaluation.

Final Recommendation

Best Overall: Perplexity

Perplexity provided the best balance between clarity, simplicity, instruction-following, and practical usefulness. It delivered an accurate explanation in plain language while staying within the requested constraints.

Gemini was a very close second because of its excellent structure, while Claude excelled in readability but included more detail than requested. GitHub Copilot correctly identified the request as outside its primary purpose, demonstrating good safety and scope awareness but not fulfilling the user's task.

Key Takeaways

This evaluation demonstrated that different AI models have unique strengths depending on the task. While all models except GitHub Copilot successfully explained Bitcoin, Perplexity produced the strongest overall response by balancing clarity, simplicity, and adherence to the prompt. The exercise also highlighted the importance of structured evaluation methods when assessing AI-generated outputs.

Skills Demonstrated

AI Model Evaluation

Prompt Engineering

Comparative Analysis

Technical Writing

Documentation

Critical Thinking

Quality Assessment

AI-Assisted Research

Communication